

Seaglider fundamentals

Principles of operation

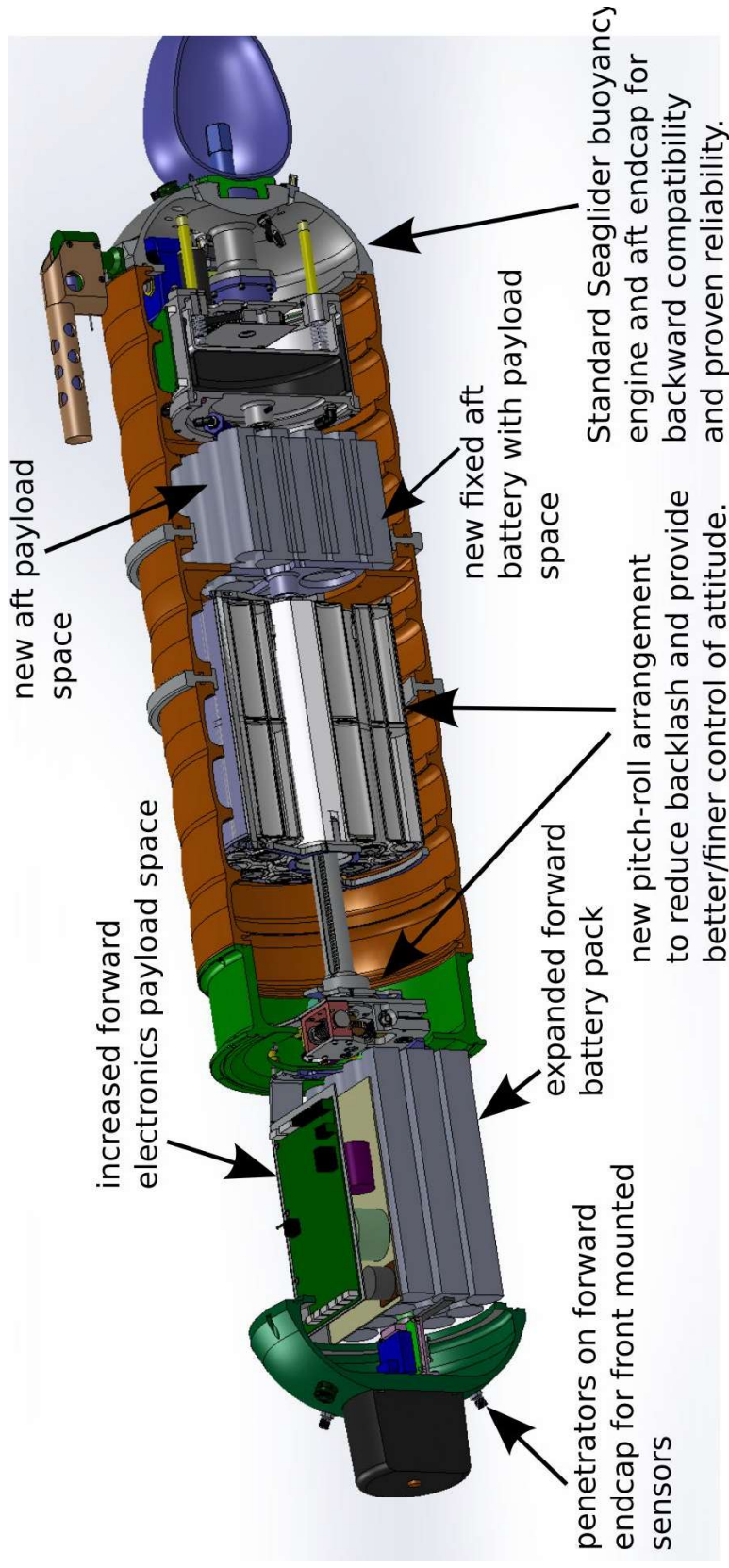
- Seagliders are buoyancy-driven AUVs: they move through the water without an active thruster and are thus extremely energy efficient.
- Gliders change their buoyancy to move up and down in the water column, much like an Argo float profiles up and down. Vertical speed is ~ 10 cm/s.
- With a glider, some vertical motion is translated into horizontal motion by the wings
 - glide slopes from $\sim 1:1$ to $\sim 1:5$, horizontal speed ~ 25 cm/s.
- The glider can control its pitch and roll by shifting its large internal battery
- It uses roll to bank and turn so that it can maintain a specified heading.
- Seaglider is designed for max energy efficiency
 - Main energy draw is buoyancy pump ($\sim 50\%$)

Seaglider components

(Not shown: antenna & mast)



SGX: the newest generation of Seaglider



Buoyancy

- For the glider to move vertically, its density must change:
 - If the glider is more dense than the surrounding water, it sinks (dives)
 - If the glider is less dense than the surrounding water, it rises (climbs)
- The glider must be ballasted or configured initially so that it is very close to neutrally buoyant in the ocean water that it will operate in.
- The glider changes its density (and thereby its buoyancy) by changing its volume while keeping its mass constant.
- It is a constant mass variable volume vehicle, unlike submarines which change their mass (by taking on or ejecting water).
- We might say the vehicle is “heavy” or “light” in a particular phase of flight, but we mean denser than surrounding water (heavy) or less dense than surrounding water (light).
- The vehicle never changes its mass while operating (we hope)

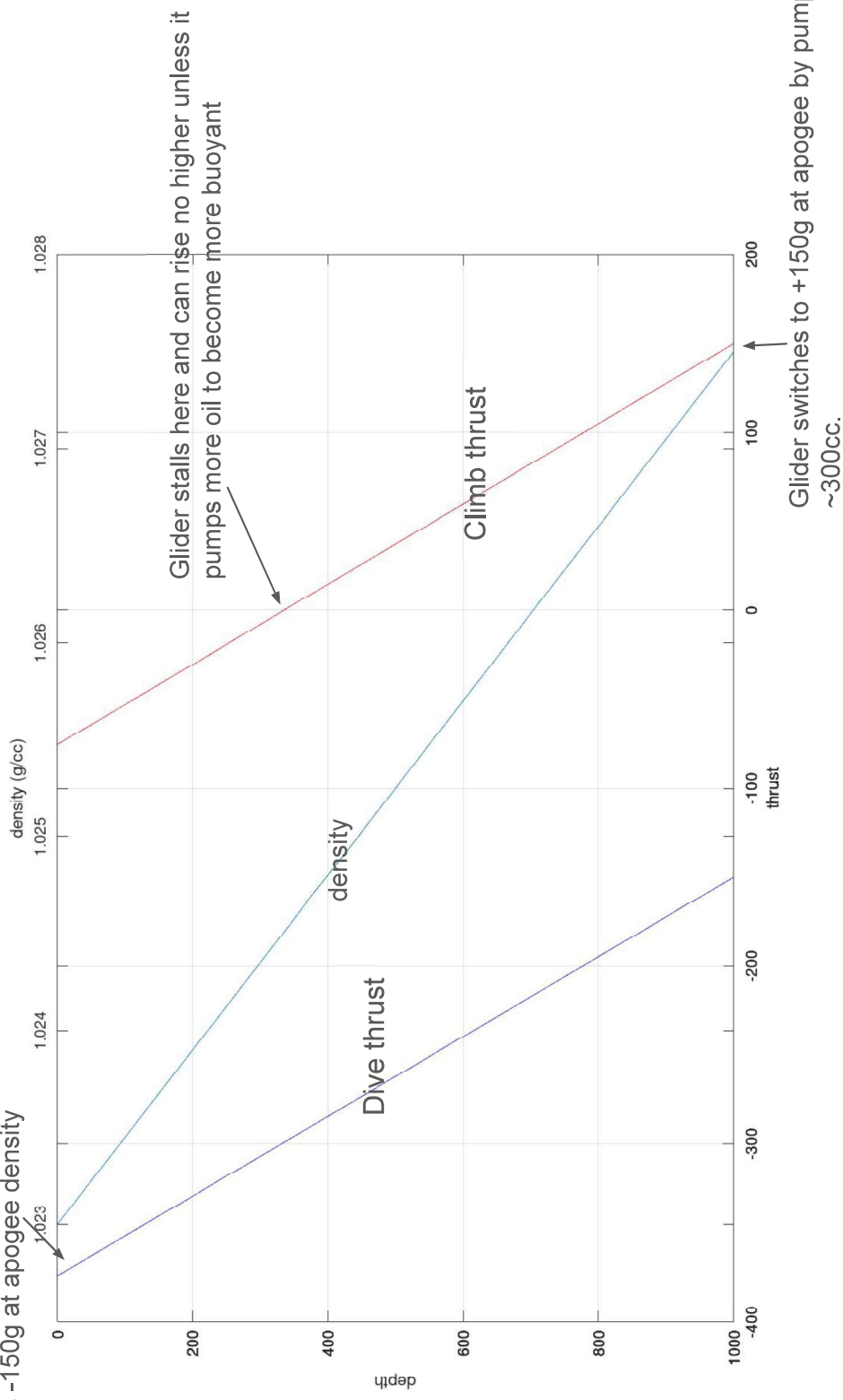
Density

- Typical 1000 m ocean density is 1027.5 kg/m^3
- Shorthand notation for density: $\sigma = (\text{density} - 1000) \text{ kg/m}^3$ – so the typical 1000 m density is 27.5 in σ units (or σ_θ since density is measured based on potential temperature θ)
- We use **potential density** (which removes the effect of pressure on density) when talking about Seaglider because Seaglider (mostly) matches the compressibility of seawater due to its hull design.
- The most important formula in glider operations:

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

Buoyancy, density and stratification - a very contrived example

Glider starts dive by bleeding to -150g thrust, but that's -150g at apogee density



Buoyancy and ballasting

The glider can be ballasted (mass changed by adding or removing lead) for

- A given range of water density
- Negative buoyancy (thrust), e.g., in strong currents
- Reserve buoyancy, e.g., in areas with low surface density like the Arctic, river plumes or Bay of Bengal (monsoon season).

Rule of thumb:

- For Seaglider SG, a ~50 cc change in glider volume produces a 1 σ_θ density change
- For SGX, a ~70 cc change in volume produces a 1 σ_θ density change

Buoyancy and ballasting

The VBD for contains around 800 cc's of moveable oil, i.e., available buoyancy (same for SG and SGX). To determine the density range (stratification) where your glider can operate:

- Start with the total buoyancy (cc's of oil) available
- Subtract the cc's of oil needed to lift the antenna out of the water
- Subtract the cc's of oil needed for negative buoyancy (thrust), usually specified at the deepest/densest water to be encountered).
- Divide the remaining cc's of oil by the rule of thumb (50 or $70 \text{ cc}/\sigma_\theta$):

Total VBD available	800 cc
Positive buoyancy required to expose antenna	(150 cc)
VBD remaining	650 cc
Negative thrust in densest water	(250 cc)
VBD remaining	400 cc
equivalent σ_T units of stratification (SGX)	5.5
equivalent σ_T units of stratification (SG)	8

Variable buoyancy device (VBD): drives glider's vertical motion

VBD moves oil between an internal reservoir that is inside the glider's pressure housing and a bladder that is external to the pressure housing, thereby changing the total volume of the glider and hence its density

Oil pumped from the reservoir to the bladder:

- Glider's total volume increases
- Glider becomes buoyant relative to the ocean around it and rises (i.e., climbs)

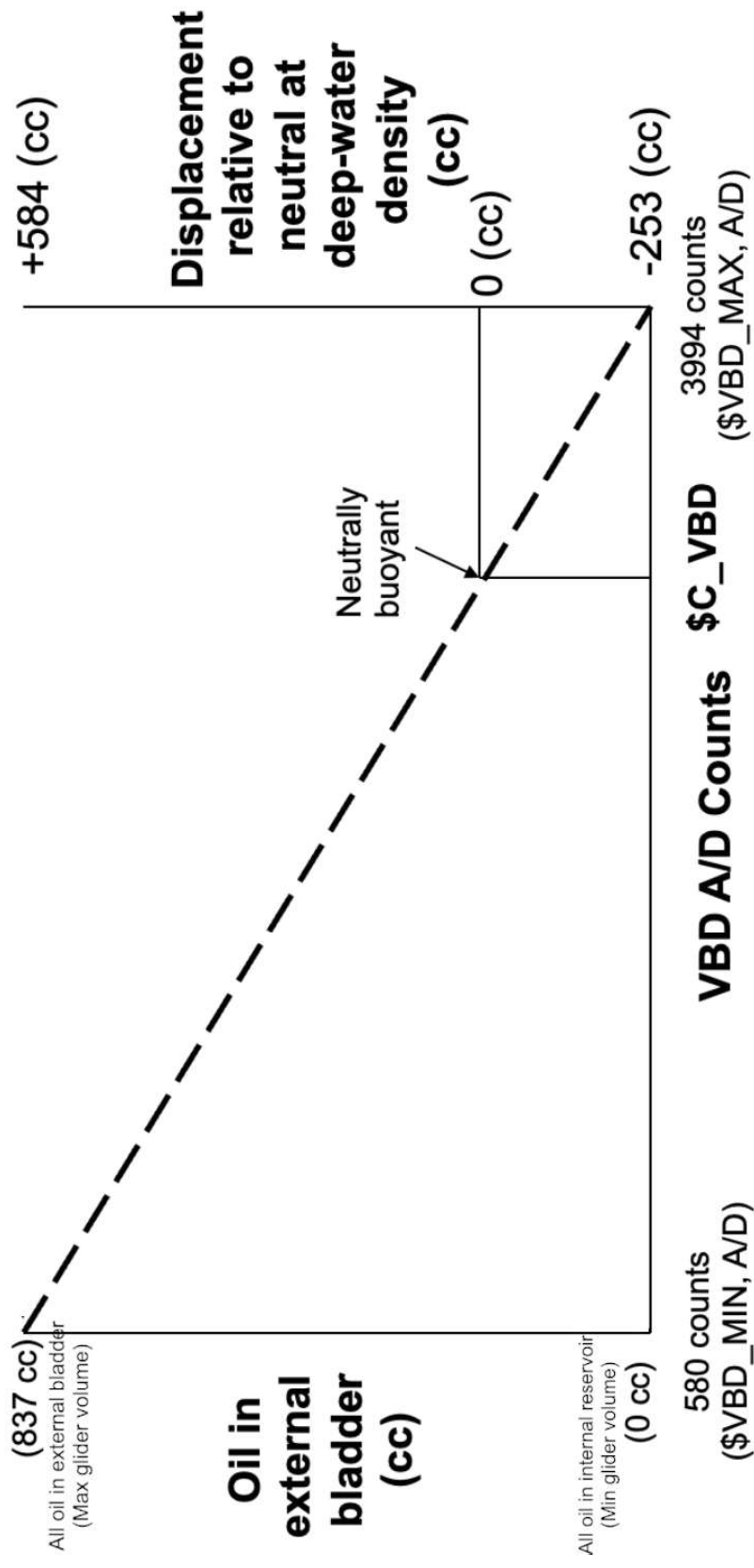
Oil bled from the bladder to the reservoir:

- Glider's total volume decreases
- Glider becomes dense relative to the ocean around it and sinks (i.e., dives)

VBD units

- For VBD there are two units that we use for talking about the control system
 - (1) In *engineering units* we talk about volume in cc's, cubic centimeters
 - This could be the maximum volume of the vehicle (which we'll call volmax below)
 - Or it could be the differential volume as measured by the oil in (or out of the bladder) relative to the volume at which the glider is neutrally buoyant.
 - (2) At the lowest level, we use A-to-D counts. This is the system on the glider that measures the control positions. It is a 12-bit value so for all systems ranges from 0 to 4095, though in practice there are software and hardware stops that put the ranges from 100s to 3900s.
- For the VBD system, low AD counts correspond to a full bladder. Higher AD counts are an emptier bladder.
- The conversion is set by the physical travel of the linear potentiometers and geometry of the internal reservoir where the volume is measured (so large AD counts is a full internal reservoir). There are 4.0767 AD counts / cc of oil for SG and SGX.
- These same concepts of engineering units and AD counts will apply to pitch and roll as well.
- Most piloting is done in AD counts.
- Most planning / analysis is done in engineering units.
- A pilot must be familiar and comfortable translating back and forth between the two concepts.

VBD oil vs AD counts (example values)



Definition of important VBD variables

Volmax: maximum volume of the glider (i.e., volume when all moveable oil is in the external bladder), in cc's. Getting an accurate estimate of volmax is key to ballasting. We get an initial estimate of volmax in a tank, and a more refined estimate during the test deployment. Volmax is a constant for a given glider configuration (external payload and VBD settings).

Neutral volume: This is the volume of the glider including external bladder position at which the glider is neutrally buoyant at the desired apogee density. By definitions below, the bladder position at this point is 0 cc and **\$C_VBD** AD counts.

Thrust: buoyancy in grams relative to the surrounding water. Negative thrust means the glider is “heavy” or denser than the surrounding water and will descend. Sometimes also described in cc of oil relative to **\$C_VBD**.

Target thrust (\$MAX_BUOY): This is the desired amount of negative thrust that we want to have at apogee so that we can maintain speed throughout the entire dive.

\$C_VBD: The position of the oil reservoir in AD counts that defines where the glider has zero thrust (is neutrally buoyant) at apogee density.

Apogee density (\$RHO): Reference density in kg/m^3 . Typically set to the most dense water expected to be seen.

\$VBD_MIN, \$VBD_MAX: AD limits of the VBD system (**\$VBD_MIN** means pumped to max, i.e., bladder is full of oil; **\$VBD_MAX** means bled to minimum, i.e., internal reservoir full of oil)

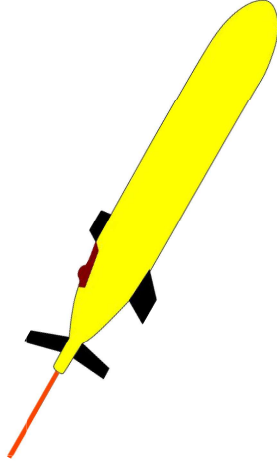
Working with VBD values

1. Target thrust fixes **\$C_VBD** because we never need to get heavier than that thrust, so we can set **\$C_VBD** such that when bled to **\$VBD_MAX** (minimum buoyancy) we will be at target thrust:
$$C_VBD = VBD_MAX - (thrust/\$RHO)/VBD_CNV$$
 2. Vol0, glider total neutral volume, is defined by glider mass and target apogee density:
$$Vol0 = mass/\$RHO$$
 3. Now we know how positive we can get: it's the difference between neutrally buoyant at **\$C_VBD** (when glider has total volume vol0) and having maximum buoyancy when pumped to **\$VBD_MIN**:
$$volmax = vol0 + (VBD_MIN - C_VBD)*VBD_CNV$$
 4. If planning or ballasting, we can rearrange (2) and (3) to solve for mass to get desired total mass of glider:
$$mass = RHO*(volmax - (VBD_MIN - C_VBD)*VBD_CNV)$$
 5. We can also confirm that the glider will be able to surface and call - we need to be able to lose 150cc (the volume of the antenna) and still be buoyant in the lowest density surface water that we expect to see:
$$RHO_min = mass / (Volmax - 150)$$
- * **\$VBD_CNV** = cc's of oil per AD count = -0.2453 (constant, same for SG and SGX)

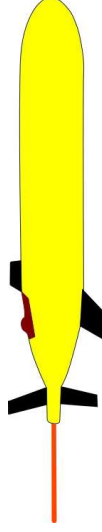
Mass shifter: controls glider's pitch and roll

The mass shifter controls the glider's pitch and roll by using electric motors to move the battery pack and a brass weight (~13 kg total mass).

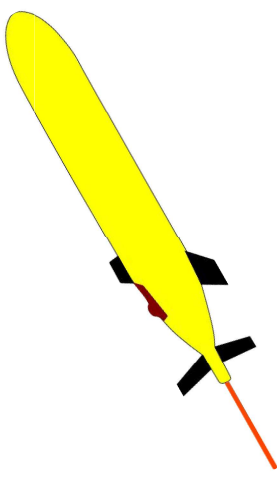
Shifting the mass forward/aft pitches the glider's nose down/up:



Mass is shifted forward
Glider's nose points down
AD counts low



Mass is between fore/aft
Glider flat
AD counts = **\$C_PITCH**

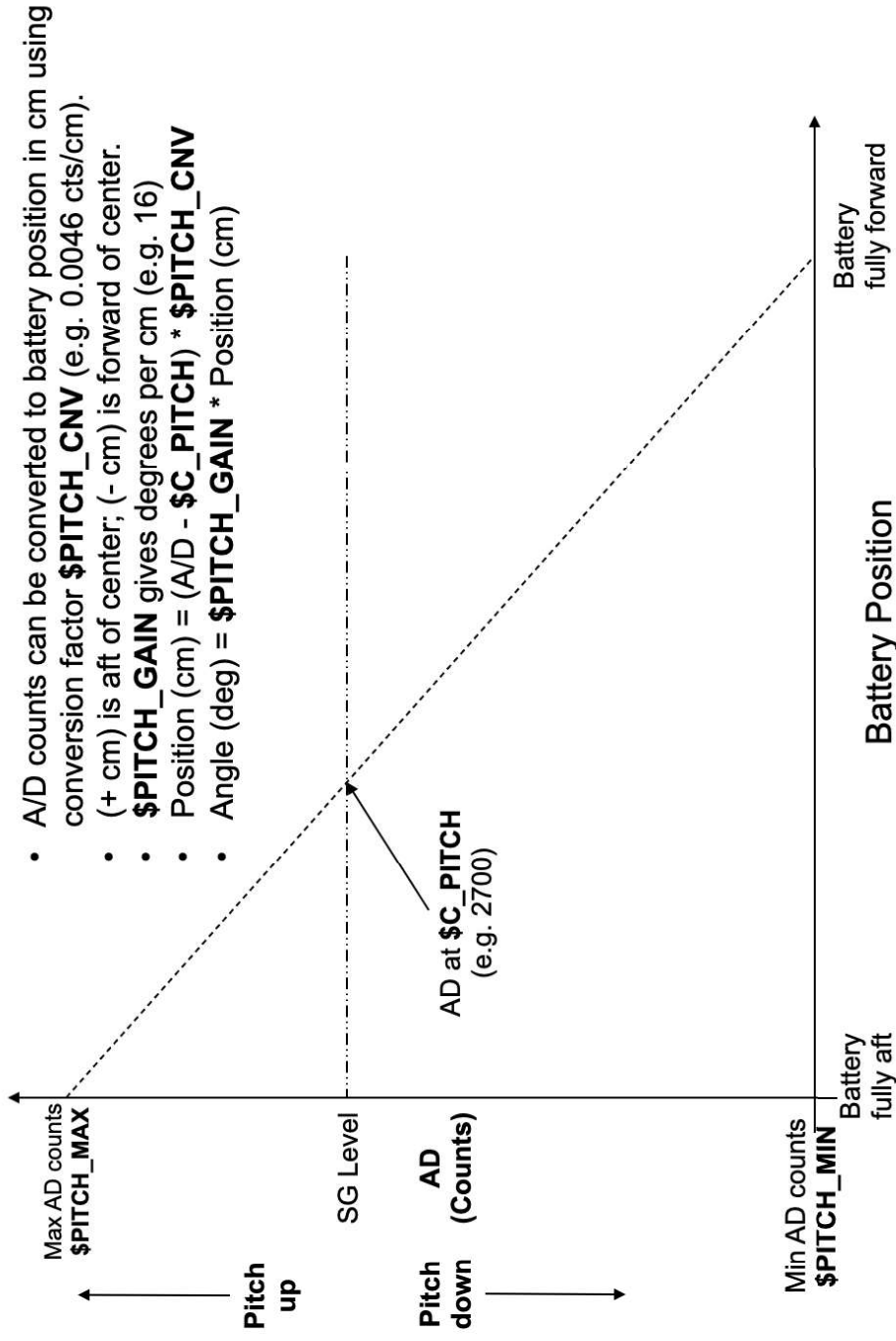


Mass is shifted aft
Glider's nose points up
AD counts high

Pitch units

- In *engineering units* we talk about the battery position along the lead screw in centimeters (cm)
- For the pitch system low AD counts correspond to the mass fully forward, glider pitched down. Higher AD counts are pitch mass aft, glider pitched up.
- The conversion (**\$PITCH_CNV**) is set by the physical travel of the lead screw and the multi-turn rotary potentiometer that measures the position of the battery along the lead screw. **\$PITCH_CNV** is different for SG and SGX but otherwise does not vary from glider to glider - it is constant for a given glider type.
- Pilots set the center (**\$C_PITCH**) in AD counts, but pitch also has a **\$PITCH_GAIN** parameter that is based on engineering units (degrees pitch / centimeter of mass movement) to describe how much the glider's pitch changes when the battery position changes.
- SGX tends to have lower gains - small battery moves produce small pitch changes. SGs are more "tippy": small battery moves produce larger pitch changes. .

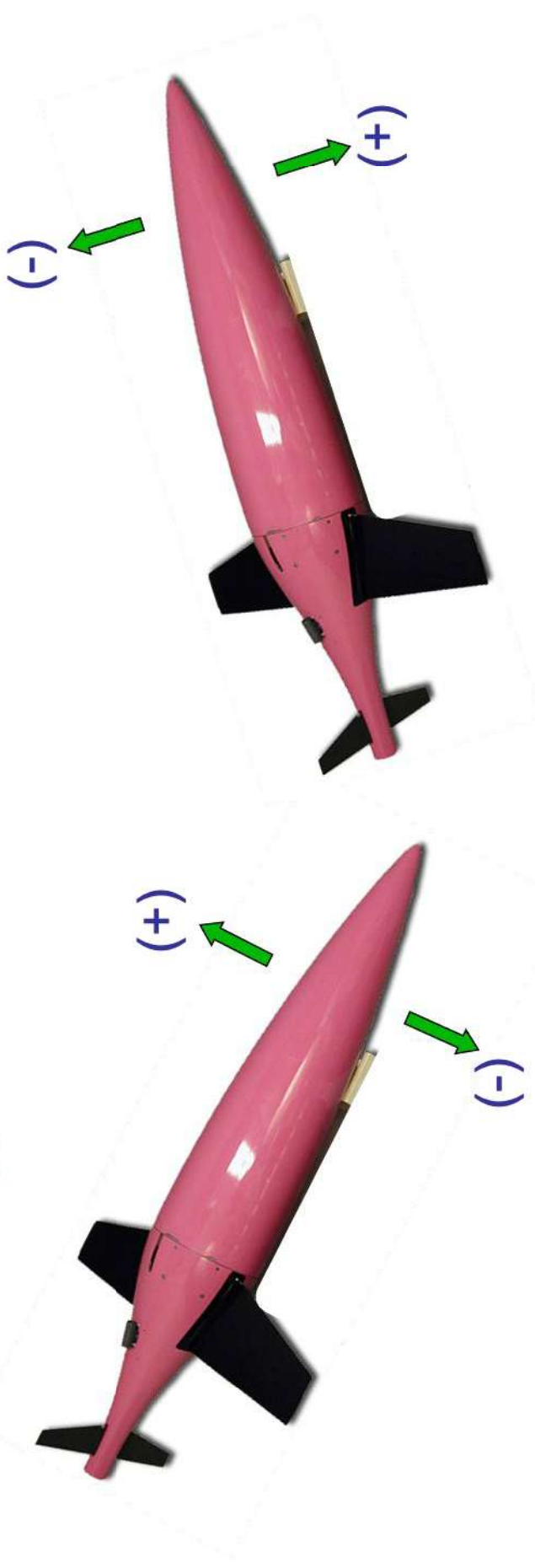
Pitch relationship to AD counts



Mass shifter: roll

- The mass shifter rotates the battery pack about the center axis of the glider
- This changes its center of gravity, making the glider roll and thereby turn
- The glider always rolls in the opposite direction as the mass shifter
- However, the direction of the turn depends on whether the glider is climbing or diving

Glider turns **opposite** on dive and climb relative to roll direction.

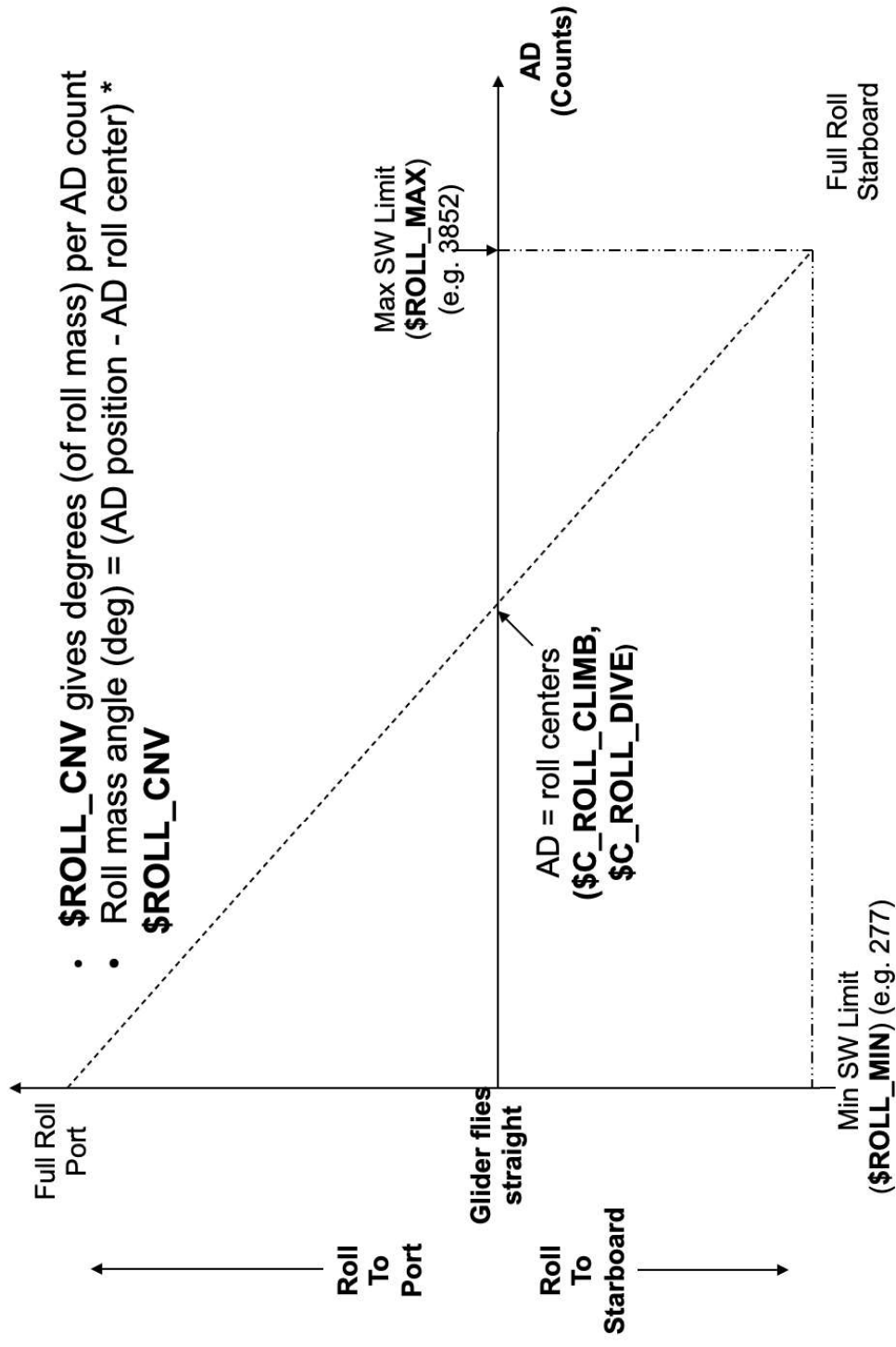


- \$C_ROLL_DIVE is to make **flight straight** during dive phase.
- \$C_ROLL_CLIMB is to make **flight straight** during climb phase.
- Smaller A/D shifts mass to port, glider rolls port
- Port roll on dive *turns right* (starboard), port roll on climb *turns left* (port)
- Larger A/D shifts mass to starboard, glider rolls starboard
- Starboard roll on dive turns left (port), starboard roll on climb turns right (starboard)

Roll units

- In *engineering units* we talk about the battery rotation around a central axis in degrees.
- Roll is a bang-bang controller - we typically apply full roll (moving the battery as far as it will go to one side or the other) - so we do not have a gain setting.
- For roll, we tune the centers in AD counts (**\$C_ROLL_DIVE**, **\$C_ROLL_CLIMB**) to specify the rotational position that best achieves straight flight when we're not trying to turn.

Roll relationship to AD counts



Recap of control concepts for buoyancy, pitch, and roll

- Electric motors drive the VBD and mass shifter systems
- Potentiometers (“pots”) are used to measure the position of each of these systems at any given time
- Potentiometers have a range of 0 to 4095 analog-to-digital (AD) counts
- AD counts are converted to physical displacements – cc’s of oil (for the VBD), cm of travel (pitch), degrees of roll (of the battery) (roll)
- Conversions are based on mechanical design of the system and do not change over time (i.e., the pilot shouldn’t change these)

Sub-system	Conversion factor	Parameter name
VBD	-0.2453 cc/AD (same for SG and SGX)	\$VBD_CNV
Pitch	0.00313 cm/AD (for SG) 0.00413 cm/AD (for SGX)	\$PITCH_CNV
Roll	0.028270 degrees/AD (for SG) 0.054945 degrees/AD (for SGX)	\$ROLL_CNV

Recap A/D counts for control of buoyancy, pitch, and roll

Sub-system	A/D counts	
	Low	High
VBD	Oil in bladder More buoyant Glider climbs	Oil in reservoir Less buoyant Glider dives
Pitch	Mass shifter forward Glider pitches down	Mass shifter aft Glider pitches up
Roll	Center of mass shifts to port (negative degrees) Glider rolls to port	Center of mass shifts to starboard (positive degrees) Glider rolls to starboard

Recap centers of buoyancy, pitch and roll

- The mass shifter and VBD have center points that are the neutral position for that sub-system (expressed in A/D counts)
- The pilot adjusts the centers to control the glider.
- Seaglidors are asymmetrical, so the roll centers are generally different when the glider is diving versus climbing

Sub-system	Parameter describing the center point	Glider position at center point
VBD	\$C_VBD	Neutrally buoyant at apogee
Pitch	\$C_PITCH	Flat
Roll	\$C_ROLL_DIVE \$C_ROLL_CLIMB	Flying straight (not turning) during dive or climb

Seaglider payload